

A decade from now, in a future characterised by continued growth, there is a machine related to robots that evokes a feeling of hope.

General idea

We would create empathetic robots that could be placed in the homes of elderly patients/clients. The robots' role would be a help system, companion and entertainment simultaneously as it is able to collect medical/health data for use by health professionals. There is ability for remote access for doctors/nurses/family members to be informed and in contact with the customer via robot assistance. The robot is able to connect to existing services like e.g. wearables (tracking health data), food delivery services, entertainment services or alarm systems. Every customer case is different and therefore the product would consist of stackable services that are personalized packages for the customers needs. Important when dealing with the elderly customers is keeping track of location and condition of the client. The robot is therefore able to call health professionals for assistance when necessary and communicates to the customer by talking. It keeps track of the important things for the customer like medication, appointments and such and makes sure for dementia customers that the medication is not repeated too often. Simple companionship and conversation with clients is an important part of this assisting robot, it is able to read the latest news or books and recommend and remind about upcoming tv programs. Gaming that activates brains is the robots forte like Alias where descriptioning a word and guessing what it is about, possibility to do the same with sounds and music. Also normal phone functions exist when typing with the fun starts to demand the robot assist with calling relatives and family. When the client starts losing important senses the robot gives haptic replies to warn and to wake attention. It has a camera system that allows distance helping and a human evaluation of the situation at hands. First aid functions like calling for help or functioning as a defibrillator are also available as add ons for the robots functions.

Cases:

Basic level care-robot for nursing homes (Hektor)

The basic level of the care-robot for nursing homes is designed to assist residents with their daily routines, provide companionship, track their health, and help residents with daily tasks. Each robot would serve multiple residents within the nursing home but would not be able to help the residents with anything requiring extensive robotic interaction. This means that the

robot would not be able to perform any complex physical tasks but instead would function as more of an assistant than a fully complete nursing-robot.

So the robot would have 3 main functionalities for residents.

1. *Empathetic companionship*, where it would act as an Alexa-like assistant, engaging in simple conversation, reminding residents about their appointments, playing games and providing entertainment such as guiding residents through exercises.
2. *Health monitoring*, which would be data driven reminders such as reminding users to drink, eat and take their medication based on the real time data. This data would also allow doctors and healthcare professionals to make more accurate prognoses about the users health.
3. *Emergency alerts and assistance*, data driven emergency alerts where if for example the robot detects a fall, it notifies healthcare personnel, or if possible it assists the user in getting up safely. If there is a sudden health decline of a user and no healthcare personnel are present, the robot could automatically alert an ambulance and be remote accessed by an healthcare professional who could try to assess the situation.

Advanced level care-robot for nursing homes (Hektor)

The advanced level care-robot would include all the same functionalities as the base version along with enhanced medical assistance and physical support. It can perform CPR, provide guided first aid, and detect critical health conditions in real time. It would also have a more adaptive personality, it tailors interactions to residents habits and preferences for improved engagement. Additionally, advanced robotics enable it to assist with mobility, including lifting residents and supporting movement.

Basic level care-robot for homes

This robot is rented for private use with stripped functions that allow its lower price. There is an alarm system for emergency and companion qualities found but the advanced functions are left to minimum.

Advanced level care-robot for homes

As a private helper the robot's mission is extending the ability of living in the customer's own home without the life quality deteriorating while age starts to become an advancing visible factor. The advanced level robot is equipped with the functions that allow it to assist with customers everyday needs from nutrition to living environment. There are functions for listing grocery needs and ability to order home delivered goods. The robot will maintain clean conditions in the household and require help when necessary.

Assumptions about the future:

- High demand: as the needs of elderly healthcare rise and birth rates decline, there will be an increasing demand for robotic nurses.
- Shortage of healthcare professionals: a declining population with a high amount of elderly will lead to a labour shortage in health care which leads to an increasing need for a robotic solution.
- Cost-effective: Robotic care is a more affordable alternative to human labour.
- Preference for home care: Elderly people will continue to prefer to live at home as long as possible, which would create a need for in-home robotic assistance.
- Feasibility: Technological advancements in robotics will make caregiving robots a viable and cost effective solution.
- Privacy concerns: users are fine with having wearable tech on them for data collections and are fine with usages and collection of this data, also the use of a camera does not bother the in-home users.
- Emotional comfort: Elderly users will see the robotic companion as a supportive friend rather than just a robot.

Other ideas:

A generation from now, in a future characterised by transformation, there is candy related to wealth that evokes a feeling of shock.

- 5g-connected device which gives impressions of taste. E.g. connecting to television or entertainment, integrating the sense of taste into the experience.

Two generations from now, in a future characterised by transformation, there is a document related to drones that evokes a feeling of optimism.

- Looser regulations for unmanned flight or drones over populated areas.
- Geographical surveys of forests and nature. Keeping track of wildlife, species.

A decade from now, in a future characterised by discipline, there is an instrument related to sports that evokes a feeling of weirdness.

- Sports wear that gathers real time data of the user to optimize performance / prevent injuries using AI-analytics (5g use case)?
- Using cloud based tech so coaches and users can get an insight on the data.

A generation from now, in a future characterised by discipline, there is an event related to drones that evokes a feeling of serenity.

- Fireworks are replaced by drone light shows.